

Coupled Cluster Gradients on One Page

What to be able to write down and what to be able to answer. Everything here is derived in the Coupled Cluster Gradients notes; the equation numbers in brackets point there.

Notation. $|0\rangle$ is the Hartree-Fock reference, $|\mu\rangle, |\nu\rangle$ are the excited determinants inside the truncation, $\tau_\mu|0\rangle = |\mu\rangle$, $\tilde{H} = e^{-T}He^T$, x is a nuclear coordinate, A^x a full derivative and $A^{(x)}$ a skeleton derivative with unperturbed orbitals. Occupied i, j , virtual a, b , general p, q .

THE FOUR EQUATIONS

$$\mathcal{L}(t, \lambda, x) = \langle 0 | \tilde{H} | 0 \rangle + \sum_{\mu} \lambda_{\mu} \langle \mu | \tilde{H} | 0 \rangle = \langle 0 | (1 + \Lambda) \tilde{H} | 0 \rangle, \quad \Lambda = \sum_{\mu} \lambda_{\mu} \tau_{\mu}^{\dagger}. \quad (1)$$

THE LAGRANGIAN [7]. The energy plus the amplitude equations times multipliers. It equals E whenever the amplitude equations hold, and unlike $E(t)$ it can be made stationary in every parameter: in λ by the amplitude equations, in t by choosing λ .

$$\langle 0 | (1 + \Lambda) (\tilde{H} - E) | \nu \rangle = 0 \text{ for all } \nu \quad \Longleftrightarrow \quad \mathbf{J}^T \boldsymbol{\lambda} = -\frac{\partial E}{\partial \mathbf{t}}, \quad J_{\mu\nu} = \frac{\partial \Omega_{\mu}}{\partial t_{\nu}}. \quad (2)$$

THE Λ EQUATIONS [10, 11]. The condition $\partial \mathcal{L} / \partial t = 0$. Linear in λ , with the transposed amplitude Jacobian; $\langle 0 | (1 + \Lambda)$ is the left eigenvector of the non-Hermitian $\tilde{H}$. Contains no perturbation, so it is solved once, at about the cost of one CCSD.

$$\frac{dE}{dx} = \langle 0 | (1 + \Lambda) \tilde{H}^x | 0 \rangle = \sum_{pq} \gamma_{pq} h_{pq}^x + \frac{1}{4} \sum_{pqrs} \Gamma_{pqrs} \langle pq || rs \rangle^x, \quad \gamma_{pq} = \langle 0 | (1 + \Lambda) e^{-T} a_p^{\dagger} a_q e^T | 0 \rangle. \quad (3)$$

THE GRADIENT [12, 15, 16]. A Hellmann-Feynman-like expectation value with a different bra and ket. No dt/dx appears: that term multiplies $\partial \mathcal{L} / \partial t$, which (2) made zero. The densities γ and Γ are built once from t and λ ; the perturbation enters only through the derivative integrals.

$$\mathbf{A}^T \mathbf{z} = \mathbf{X}, \quad X_{ai} \propto F_{ai}^{\text{gen}} - F_{ia}^{\text{gen}}, \quad \frac{dE}{dx} = \sum_{\mu\nu} D_{\mu\nu} h_{\mu\nu}^x + \frac{1}{2} \sum_{\mu\nu\lambda\sigma} \Gamma_{\mu\nu\lambda\sigma}^{\text{eff}} (\mu\nu | \lambda\sigma)^x - \sum_{\mu\nu} W_{\mu\nu} S_{\mu\nu}^x + \frac{dV_{nn}}{dx}. \quad (4)$$

THE Z-VECTOR AND THE WORKING FORMULA [22–24]. The orbitals move with the nuclei, and E is not stationary to orbital rotations, so the orbital response $\mathbf{U}^x$ would be needed for every x . One solve with the transposed orbital Hessian $\mathbf{A}$ against the CC orbital gradient $\mathbf{X}$ gives $\mathbf{z}$, which folds into the relaxed density $\mathbf{D}$, the effective two-particle density and the energy-weighted density $\mathbf{W}$, and the gradient takes the shape of a Hartree-Fock gradient.

THE RECIPE, IN THE ORDER A CODE RUNS IT

Hartree-Fock $\rightarrow$ CCSD amplitudes $t \rightarrow \Lambda$ equations (2) for $\lambda \rightarrow$ unrelaxed densities $\gamma, \Gamma \rightarrow$ generalized Fock matrix and orbital gradient $\mathbf{X} \rightarrow$ Z-vector equation (4) for $\mathbf{z} \rightarrow$ relaxed $\mathbf{D}, \mathbf{\Gamma}^{\text{eff}}, \mathbf{W}$ back-transformed to the atomic-orbital basis $\rightarrow$ contract with derivative integrals for every coordinate. Two iterative solves of CCSD size and one of CPHF size, whatever the number of atoms; central differences would need $6N_{\text{at}}$ energies.

THE FOUR QUESTIONS, WITH ANSWERS

1 Why do you need Λ at all?

Because the CC energy is not variational in the amplitudes: $\partial E / \partial t \neq 0$, so the chain rule leaves $\sum_{\mu} (\partial E / \partial t_{\mu}) (dt_{\mu} / dx)$ in the gradient. The multipliers λ are chosen so that the Lagrangian, which equals the energy, has no first-order dependence on t . Equivalently, $\tilde{H}$ is not Hermitian, so its left eigenvector $\langle 0 | (1 + \Lambda)$ must be found separately from the ket $e^T | 0 \rangle$.

3 What does the Z-vector equation solve for, and what does it replace?

It solves $\mathbf{A}^T \mathbf{z} = \mathbf{X}$ for $\mathbf{z}$, the Lagrange multipliers of the Brillouin conditions, with $\mathbf{A}$ the coupled-perturbed Hartree-Fock orbital Hessian and $\mathbf{X}$ the CC orbital gradient. It replaces solving $\mathbf{A} \mathbf{U}^x = \mathbf{B}^x$ once per coordinate: $\mathbf{X}^T \mathbf{A}^{-1} \mathbf{B}^x$ becomes $\mathbf{z}^T \mathbf{B}^x$, one solve and cheap contractions, and $\mathbf{z}$ ends up inside the relaxed and energy-weighted densities of (4).

2 Why is there no need to solve for the amplitude derivatives?

Because $\mathcal{L}$ is stationary in t . A first-order change in t changes $\mathcal{L}$ only at second order, so dt/dx drops out of a first derivative. This is the $2n + 1$ rule: wavefunction parameters to order n give energies to order $2n + 1$. Solving for dt/dx instead would mean one Jacobian solve per coordinate, $3N_{\text{at}}$ of them, each the cost of a CCSD.

4 Why are the Λ equations solved once rather than once per nuclear coordinate?

Because they do not contain the perturbation. Equation (2) is $\partial \mathcal{L} / \partial t = 0$ at the reference geometry and depends only on $\tilde{H}$ and t . The same holds for $\mathbf{z}$. The coordinate x enters only through the derivative integrals in the final contraction of (3) and (4), which is why an analytic gradient costs a constant number of energies, independent of the number of atoms.

THREE TRAPS

- Λ is not $T^{\dagger}$. They agree to first order in perturbation theory and differ beyond it; only the true Λ makes $\mathcal{L}$ stationary, and an approximate one gives forces that are not the derivative of any energy.
- Converge t and λ more tightly for a gradient than for an energy. The energy error is second order in the residuals, the gradient error is first order.
- An unrelaxed property stops after the densities of (3); a relaxed one adds the Z-vector. For nuclear gradients there is no choice, the basis moves with the nuclei and relaxation is mandatory.

Bracketed numbers are equation numbers in the Coupled Cluster Gradients notes. Spin-orbital conventions follow Shavitt and Bartlett.